

Synopsis Abstract

- One Nation One Challan is a unified centralised system aimed at streamlining the management of traffic violation fines and penalties across India, ensuring uniformity for challan generation and payment.
- A robust grievance system would provide a platform for citizens to raise concerns, report discrepancies, or seek clarifications regarding traffic violation penalties imposed.
- Our project aims to establish an online grievance portal that provides speedy issue resolution, a user-friendly interface, and awareness among citizens.

Our Project comprises ~

- 1) Implementing an AI Chatbot for the grievance lodging system improves efficiency and offers prompt support for resolving issues.
- 2) Centralised System for efficient management of diverse challan grievances across various departments.
- 3) Challans generated Analytics facilitates data-driven decision-making and process optimization.
- 4) Real-time grievance tracking ensures prompt and effective issue resolution, fostering contentment among users.
- 5) Notifications and alerts boost transparency and engagement by providing real-time grievance updates.
- 6) Citizen feedback to the National Informatics Center facilitates continuous improvements and user-centric enhancements.
- 7) Analysing frequent complaints enables proactive improvements, enhancing the overall efficiency and user experience.
- 8) Comprehensive information on One Nation One Challan fosters awareness, participation and compliance.

Literature Review

- 1) Absence of an AI-based grievance lodging system within the existing framework.
- 2) The existing system does not align with the One Nation One Challan policy.

- 3) Opaqueness in the resolution progress due to lack of timely alerts or notifications.
- 4) Fails to incorporate Frequent Complaints Analysis.
- 5) Challan analysis is not provided. (Age group, Paid/Unpaid, Std/Non-Std NPs, Gender, Vehicle Density, challan Generated density, Old/New NPs, Challan Categories)
- 6) Restricted language choices available.
- 7) Citizen Feedback is conspicuously absent.
- 8) Inconsistent policy implementation across different regions.
- 9) Lack of AI-powered customer support system.

Idea Approach

Dual Application Platform :

- 1) Web Application : A web application for the Customer Grievance Portal allows easy access to all features and resolve issues online, promoting awareness among citizens.
- 2) Mobile applications : A mobile app offers convenient access to all features, reaching more users, particularly those without access to laptops or computers, facilitating online, speedy grievance redressal for citizens.

AI Based Features :

- 1) AI based Grievance Lodging system : A software integrated with Artificial Intelligence algorithms and Natural Language Processing capabilities, enabling automated handling and processing of user complaints and issues.
- 2) AI based Helpdesk : An automated support system incorporating Artificial Intelligence technologies such as Natural Language Processing and Machine Learning to swiftly and effectively address user queries and concerns.

Analysis Features (For the POLICE DEPARTMENT) :

- 1) Generated Challan Analysis : Providing comprehensive analytics utilising statistical and computational techniques enhancing operational efficiencies.

- 2) Frequently Lodged Complaints Analysis : The systematic examination and assessment of commonly raised issues, aiming to identify patterns, root causes, and trends for proactive improvements.

Better User Experience Features :

- 1) Grievance Progress Notifications/ Alerts : Supplying immediate status and progress notifications to users about their filed complaints, maintaining transparency and user involvement as their issues are resolved.
- 2) Feedback Mechanism (For NIC) : Structured system allowing users to provide their input, opinions, and suggestions, thus enabling continuous improvement and enhancing user-centric features and services.
- 3) Access to FAQ's : Provision of a comprehensive compilation of frequently asked questions and their corresponding answers, aimed at providing users readily assistance with typical inquiries and concerns.

Road map (1)

- 1) Database Collection.
- 2) Data Cleaning and extracting key tokens.
- 3) Training the Model on the database.
- 4) Real-time Integration with Front-end.
- 5) Grievance lodging through AI Chatbot.
- 6) Grievance tracking and Progress Notifications.

Road map (2)

Tools and Tech Used

- 1) MySQL : Used to store and manage the vast amount of data for efficient data retrieval and analysis.
- 2) Flask API : Serves as the backend framework for handling requests and responses, enabling seamless interaction with database and functionality.

- 3) React : Javascript library for creating dynamic and interactive UI featuring AI chatbot and analysis.
- 4) Python : High-Level programming language with Libraries like TensorFlow, Pandas, Keras, NumPy used for incorporating features data processing, analysis, etc.
- 5) Natural Language Processing : AI powered Chatbot to facilitate automatic processing of citizen grievances for sentiment, categorization, and responses.

Challenges

- 1) Collection of Dataset : Gathering and managing data from diverse sources like CCTV cameras, police databases, and NIC databases can be complex and resource-intensive.
- 2) Limited Infrastructure : Resource Constraints pose a challenge due to limited hardware, network, and server capacity, which might lead to False challan generation.
- 3) Computational Power : Requirement for robust and high-performing computing resources to handle complex data analysis, AI-driven functionalities, and real-time processing.
- 4) Security and Data Privacy : Implementing robust security measures to protect sensitive citizen data and payment information.
- 5) Scalability : Ensuring that the system can handle a large number of users and requests while maintaining performance.

Possible Outcomes

- 1) Enhanced user experience and satisfaction through streamlined and efficient grievance resolution processes.
- 2) Improved transparency and accountability in the management of traffic violation fines and penalties.
- 3) Reduction of Falsely Generated Challans through improved data accuracy.

- 4) Increased public awareness and participation in the One Nation One Challan initiative.
- 5) Data-driven insights for informed decision-making and policy formulation.
- 6) Strengthened trust and credibility in the governance system through effective communication and user engagement.
- 7) Optimal resource utilisation and cost-efficiency in managing challan grievances.
- 8) A scalable and adaptable platform for potential future integrations and enhancements in the governance ecosystem.

Work Done till date

- 1) User Research.
- 2) Literature Review + Database.
- 3) SWOT Analysis.
- 4) Idea Approach.
- 5) Architect.
- 6) Flowgram.

SWOT Analysis

Strengths:

- 1) Centralised and streamlined management of challans.
- 2) Improved efficiency in grievance resolution with AI integration.
- 3) Efficient Issue Tracking reduces response time.
- 4) Real-time Grievance Progress Notifications / Alerts.
- 5) Valuable feedback collection for continuous improvement.
- 6) Utilises data for analytics, improving services and evaluations.
- 7) Establishes transparency through open communication and resolutions.
- 8) User-friendly interface for enhanced accessibility and user adoption.

Weaknesses :

- 1) Data collection complexity due to diverse sources.
- 2) Infrastructure constraints may lead to false challan generation.
- 3) High Computational Power to support complex data analysis and AI-driven functions.

Opportunities :

- 1) Integration with other government systems.
- 2) Use of analytics to identify trends and patterns.

- 3) Increased public awareness and participation through outreach initiatives
- 4) Data-driven policy enhancements for effective governance
- 5) Future scalability maintaining efficiency and performance.

Threats :

- 1) Data security and privacy concerns.